# ELG 3125 - Signals & Systems

## Midterm Review

**Q1)** Determine if the following functions are periodic or not. If yes, draw the function and then find the fundamental period.

a) $x[n] = \exp\left[j\left(\frac{3\pi}{2}n - 2\right)\right]$

$x[n] = x[n+N]$, $x[n] = e^{j\left(\frac{3\pi}{2}n - 2\right)} = e^{j\left(\frac{3\pi}{2}n\right)} e^{-j(2)}$

$x[n+N] = e^{j\left(\frac{3\pi}{2}(n+N) - 2\right)} = e^{j\left(\frac{3\pi}{2}n + \frac{3\pi}{2}N - 2\right)} = e^{j\left(\frac{3\pi}{2}\right)n} e^{j\left(\frac{3\pi}{2}\right)N} e^{-j(2)}$

$\Rightarrow e^{j\left(\frac{3\pi}{2}\right)n} e^{j\left(\frac{3\pi}{2}\right)N} e^{-j(2)} = e^{j\left(\frac{3\pi}{2}\right)n} e^{-j(2)}$

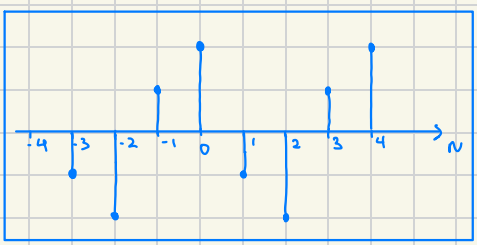
$\Rightarrow e^{j\left(\frac{3\pi}{2}\right)N} = 1$

$\Rightarrow \cos\left(\frac{3\pi}{2}N\right) + j\sin\left(\frac{3\pi}{2}N\right) = 1 + j0 \Rightarrow \cos\left(\frac{3\pi}{2}N\right) = 1 \Rightarrow \frac{3\pi}{2}N = 2\pi m \Rightarrow N = \frac{2(2)}{3}m = \frac{4}{3}m$, $m = 3 \Rightarrow \boxed{N = 4}$

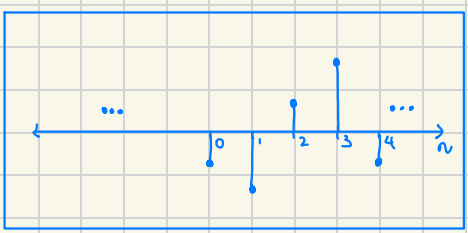
$\Rightarrow x[n]$ is periodic

$x[n] = e^{j\left(\frac{3\pi}{2}n - 2\right)} = \underbrace{\cos\left(\frac{3\pi}{2}n - 2\right)}_{\text{Real}} + \underbrace{j\sin\left(\frac{3\pi}{2}n - 2\right)}_{\text{Imaginary}}$

Real:




Imaginary:




b) $x[n] = \cos\left(\left(\frac{7\pi}{5}\right)n\right)\cos\left(\left(\frac{\pi}{3}\right)n\right)$

$\cos(a)\cos(b) = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$

$\cos\left(\left(\frac{7\pi}{5}\right)n\right)\cos\left(\left(\frac{\pi}{3}\right)n\right) = \frac{1}{2}\left[\cos\left(\frac{7\pi}{5}n + \frac{\pi}{3}n\right) + \cos\left(\frac{7\pi}{5}n - \frac{\pi}{3}n\right)\right] = \frac{1}{2}\left[\cos\left(\frac{7\pi n(3) + 5\pi n}{15}\right) + \cos\left(\frac{7\pi n(3) - 5\pi n}{15}\right)\right]$

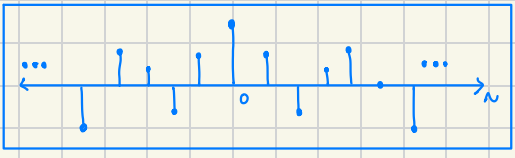
$\Rightarrow \boxed{x[n] = \frac{1}{2}\left[\cos\left(\frac{26\pi n}{15}\right) + \cos\left(\frac{16\pi n}{15}\right)\right]}$

$\frac{16\pi}{15}N_2 = 2\pi m \Rightarrow N_2 = \frac{2\pi(15)}{16\pi}m = \frac{15}{8}m$, $m = 8 \Rightarrow \boxed{N_2 = 15}$

$\frac{26\pi}{15}N_1 = 2\pi m$

$\Rightarrow N_1 = \frac{2\pi(15)}{26\pi}m = \frac{15}{13}m$, $m = 13 \Rightarrow \boxed{N_1 = 15}$

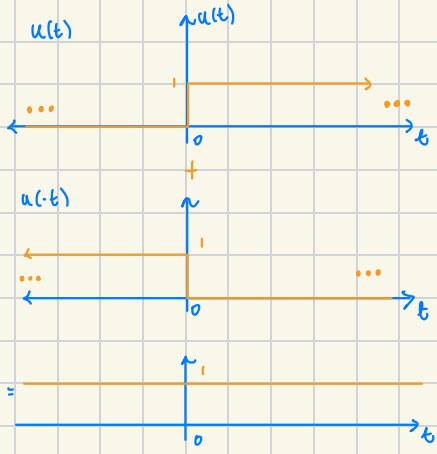
$\Rightarrow \boxed{N = 15}$




$\Rightarrow x[n]$ is periodic

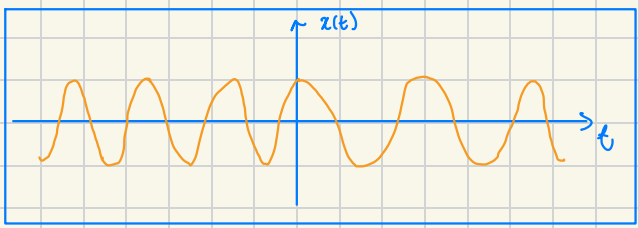
c) $x(t) = E_v[\underbrace{\cos(2\pi t)u(-t)}_{g(t)}] = EV\{g(t)\} = g_e(t)$

$g_e(t) = \frac{1}{2}[g(t)+g(-t)] = \frac{1}{2}[\cos(2\pi t)u(-t) + \cos(-2\pi t)u(-(-t))] = \frac{1}{2}[\cos(2\pi t)u(-t) + \cos(2\pi t)u(t)] = \frac{1}{2}\cos(2\pi t)[\underbrace{u(-t)+u(t)}_{=1}]$




$\Rightarrow g_e(t) = \frac{1}{2}\cos(2\pi t) \Rightarrow x(t) = \frac{1}{2}\cos(2\pi t) \rightarrow \cos(2\pi T) = 1$

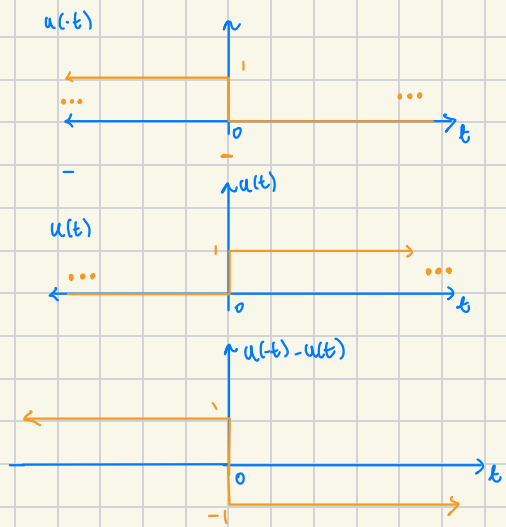
$2\pi T = 2\pi \Rightarrow \boxed{T = 1} \Rightarrow x(t)$ is periodic




d) $x(t) = Od[\underbrace{\cos(2\pi t)u(-t)}_{g(t)}] = Od\{g(t)\} = g_o(t)$, $g_o(t) = \frac{1}{2}[g(t)-g(-t)]$

$g_o(t) = \frac{1}{2}[\cos(2\pi t)u(-t) - \cos(-2\pi t)u(-(-t))] = \frac{1}{2}[\cos(2\pi t)u(-t) - \cos(2\pi t)u(t)] = \frac{1}{2}\cos(2\pi t)[u(-t)-u(t)]$

$x(t)$ not periodic because is experiences a discontinuity @ $t=0$




Q2) A discrete-time causal LTI system described by a difference equation given by: $y[n] = \frac{5}{6}y[n-1] + x[n]$

a) Find the impulse response $h[n]$.

b) Is it an invertible system? If yes, find the impulse response $h_i[n]$ of the invertible system.

c) If a signal $x[n] = u[n] - u[n-5]$ is applied to the input, find the output $y[n]$.

a) when $x[n]=\delta[n]$, $y[n]=h[n]$

$\Rightarrow h[n]=\frac{5}{6}h[n-1]+\delta[n]$

$h[0]=\frac{5}{6}h[-1]+\delta[0]=1=\left(\frac{5}{6}\right)^0$ ($h[-1]\to 0$ → LTI condition; $\delta[0]=1$)

$h[1]=\frac{5}{6}h[1-1]+\delta[1]=\frac{5}{6}h[0]=\frac{5}{6}(1)=\left(\frac{5}{6}\right)^1$

$h[2]=\frac{5}{6}h[2-1]+\delta[2]=\frac{5}{6}h[1]=\frac{5}{6}\left(\frac{5}{6}\right)=\left(\frac{5}{6}\right)^2$

$h[3]=\frac{5}{6}h[3-1]+\delta[3]=\frac{5}{6}h[2]=\frac{5}{6}\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)=\left(\frac{5}{6}\right)^3$

$\vdots$

$h[n]=\left(\frac{5}{6}\right)^n u[n]$ ↳ because system is LTI

$\Rightarrow \boxed{h[n]=\left(\frac{5}{6}\right)^n u[n]}$

b) $y[n]=\frac{5}{6}y[n-1]+x[n]$

high-school method type skii

swap $x[n]$ & $y[n]$

$\Rightarrow x[n]=\frac{5}{6}x[n-1]+y[n]$

when $x[n]=\delta[n]\Rightarrow y[n]=h_1[n]$

$\Rightarrow \delta[n]=\frac{5}{6}\delta[n-1]+h_1[n]$

$\boxed{h_1[n]=\delta[n]-\frac{5}{6}\delta[n-1]}$

$\boxed{h[n]*h_1[n]=1}$

Yes, it is invertible because $x[n]$ can be expressed uniquely in terms of $y[n]$.

c) $x[n]=u[n]-u[n-5]$

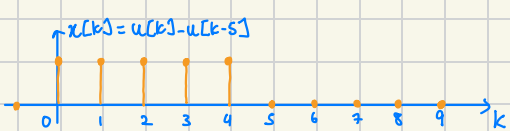
$h[n]=\left(\frac{5}{6}\right)^n u[n]$

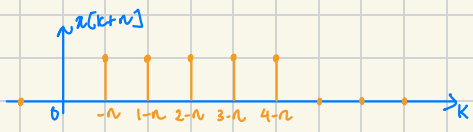
$y[n]=x[n]*h[n]=\sum_{k=-\infty}^{\infty}h[k]x[n-k]$, $x[n-k]=x[-k+n]$

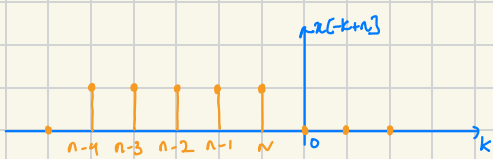


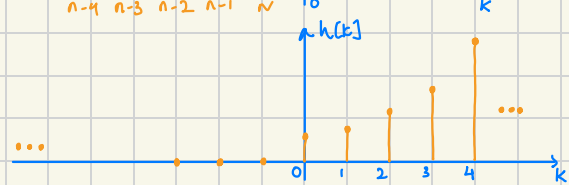






for $n<0$ : no overlap $\Rightarrow$ $\boxed{y[n]=0}$

$n\geq 0$ & $n-4<0 \Rightarrow n<4$

$\Rightarrow$ for: $0\leq n<4 \Rightarrow y[n]=\sum_{k=0}^{n} h[k]\underbrace{x[n-k]}_{=1}=\sum_{k=0}^{n}\left(\frac{5}{6}\right)^k\underbrace{u[k]}_{=1}=\sum_{k=0}^{n}\left(\frac{5}{6}\right)^k=\frac{1-(5/6)^{(n+1)}}{1-(5/6)}$

$\Rightarrow$ for $0\leq n\leq 4 \Rightarrow$ $\boxed{y[n]=\frac{1-(5/6)^{(n+1)}}{1-(5/6)}}$

$n-4>0 \Rightarrow n>4$

for $n>4 \Rightarrow y[n]=\sum_{k=n-4}^{n} h[k]\underbrace{x[n-k]}_{=1}=\sum_{k=n-4}^{n}\left(\frac{5}{6}\right)^k\underbrace{u[k]}_{=1}=\sum_{k=n-4}^{n}\left(\frac{5}{6}\right)^k=\frac{(5/6)^{(n-4)}-(5/6)^{(n+1)}}{1-(5/6)}$

$\Rightarrow$ for $n>4 \Rightarrow$ $\boxed{y[n]=\frac{(5/6)^{(n-4)}-(5/6)^{(n+1)}}{1-(5/6)}}$

final result

$$\Rightarrow y[n]=\begin{cases}0, & n<0\\ \frac{1-(5/6)^{(n+1)}}{1-(5/6)}, & 0\leq n\leq 4\\ \frac{(5/6)^{(n-4)}-(5/6)^{(n+1)}}{1-(5/6)}, & n>4\end{cases}$$

Q3) For a causal LTI system that is described by the following differential equation

$$\frac{dy(t)}{dt}+2y(t)=5x(t)$$

a) Find its impulse response $h(t)$ Note: do not use transforms.

b) If an input signal $x(t)=\delta(t)+e^{-t}u(t)$ is applied to the input of the system, find its output $y(t)$ (hint: calculate the convolution)

a) $\frac{dy(t)}{dt}+2y(t)=5x(t)$

when $x(t)=\delta(t)$, $y(t)=h(t)$

$\Rightarrow \frac{dh(t)}{dt}+2h(t)=5\delta(t)$

$h(t)=h_h(t)+\cancel{h_p(t)}^{\,0}$

Guess: $h_h(t)=Ae^{\lambda t}$, $\frac{dh_h(t)}{dt}=\lambda Ae^{\lambda t}$

$\frac{dh_h(t)}{dt}+2h_h(t)=0 \Rightarrow \lambda Ae^{\lambda t}+2Ae^{\lambda t}=0 \Rightarrow \underbrace{Ae^{\lambda t}}_{>0}(\lambda+2)=0 \Rightarrow \boxed{\lambda=-2}$

$\Rightarrow \boxed{h_h(t)=Ae^{-2t}}$

for $t<0 \Rightarrow h(t)=0$

$\frac{dh(t)}{dt} + 2h(t) = 5\delta(t) \Rightarrow dh(t) + 2h(t)dt = 5\delta(t)dt \Rightarrow \int_{0^-}^{0^+}[dh(t)+2h(t)dt] = \int_{0^-}^{0^+} 5\delta(t)dt$

(contains no delta fn)

$\Rightarrow \int_{0^-}^{0^+} dh(t) + 2\int_{0^-}^{0^+} h(t)dt = 5\int_{0^-}^{0^+}\delta(t)dt \Rightarrow h(0^+) - h(0^-) = 5(1) \Rightarrow h(0^+) = 5$ ($\int_{0^-}^{0^+}\delta(t)dt = 1$)

$h(t) = Ae^{-2t} \Rightarrow h(0^+) = Ae^{-2(0^+)} = 5 \Rightarrow \boxed{A=5}$ ($e^{-2(0^+)} = 1$)

$\Rightarrow \boxed{h(t) = 5e^{-2t}u(t)}$ scaling by $u(t)$ is important $\Rightarrow$ for causality component of the LTI system

b) $x(t) = \delta(t) + e^{-t}u(t)$

$h(t) = 5e^{-2t}u(t)$

$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} h(t-\tau)[\delta(\tau)+e^{-\tau}u(\tau)]d\tau = \underbrace{\int_{-\infty}^{\infty}\delta(\tau)h(t-\tau)d\tau}_{=y_1(t)} + \underbrace{\int_{-\infty}^{\infty} e^{-\tau}u(\tau)h(t-\tau)d\tau}_{=y_2(t)}$

$y(t) = y_1(t) + y_2(t)$

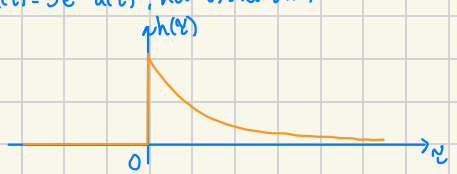
$y_1(t) = \int_{-\infty}^{\infty}\delta(\tau)h(t-\tau)d\tau = \delta(t) * h(t) = h(t)$, using delta-function property.

$\Rightarrow \boxed{y_1(t) = h(t)} \Rightarrow \boxed{y_1(t) = 5e^{-2t}u(t)}$

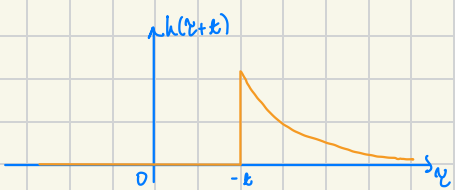
$y_2(t) = \int_{-\infty}^{\infty} e^{-\tau}\underbrace{u(\tau)}_{=1}h(t-\tau)d\tau$, $h(t) = 5e^{-2t}u(t) \Rightarrow h(t-\tau) = 5e^{-2(t-\tau)}u(t-\tau) = 5e^{-2t}e^{2\tau}u(t-\tau)$

$\Rightarrow y_2(t) = \int_{-\infty}^{\infty} e^{-\tau}5e^{-2t}e^{2\tau}\underbrace{u(t-\tau)}_{=1}d\tau = 5e^{-2t}\int_{-\infty}^{\infty} e^{\tau}d\tau \Rightarrow \boxed{y_2(t) = 5e^{-2t}\int_{-\infty}^{\infty} e^{\tau}d\tau}$

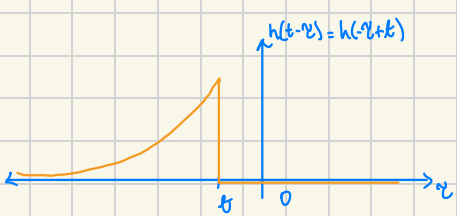
$h(\tau) = 5e^{-2\tau}u(\tau)$, $h(t-\tau) = h(-\tau+t)$

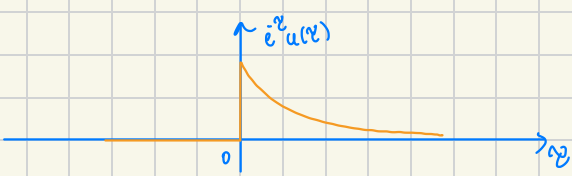



$\Rightarrow$




$\Rightarrow$



for $t<0$: no overlap $\Rightarrow$ $y(t)=0$

for $t\geq 0$: $y_2(t)=5e^{-2t}\int_{-\infty}^{\infty} e^{\tau}d\tau = 5e^{-2t}\int_0^t e^{\tau}d\tau = 5e^{-2t}\left[e^{\tau}\right]_0^t = 5e^{-2t}[e^t-e^0] = 5e^{-2t}[e^t-1] = 5(e^{-t}-e^{-2t})$

$\Rightarrow$ $y_2(t)=5(e^{-t}-e^{-2t})u(t)$ (for causality)

$y(t)=y_1(t)+y_2(t)=5e^{-2t}u(t)+5(e^{-t}-e^{-2t})u(t)=\cancel{5e^{-2t}u(t)}+5e^{-t}u(t)-\cancel{5e^{-2t}u(t)}=5e^{-t}u(t)$

$\Rightarrow$ $y(t)=5e^{-t}u(t)$ output $y(t)$ for input $x(t)=\delta(t)+e^{-t}u(t)$

Q4-1) A continous-time LTI system has the following impulse response:

$h(t)=e^{-3t}u(t+1)$

a) Is it a causal system? Justify

b) Is it a stable system? Justify

c) What is the system output when an input signal $x(t)=u(t)-u(t-5)$ is applied to the system?

a) @ $t=-1$

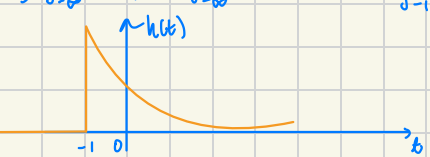
$\Rightarrow h(t)=e^{-3t}u(t+1)$

$\Rightarrow h(-1)=e^{-3(-1)}u(-1+1)=e^3\underbrace{u(0)}_{=1}=e^3\neq 0$ $\Rightarrow$ non causal

b) Stability condition

$\int_{-\infty}^{\infty}|h(t)|dt<\infty$

$\Rightarrow \int_{-\infty}^{\infty}|h(t)|dt=\int_{-\infty}^{\infty}|e^{-3t}u(t+1)|dt=\int_{-1}^{\infty}|e^{-3t}\underbrace{u(t+1)}_{=1}|dt=\int_{-1}^{\infty}\underbrace{|e^{-3t}|}_{>0}dt=\int_{-1}^{\infty}e^{-3t}dt$, $[-3t]'=-3 \Rightarrow 1=-\frac{1}{3}[-3t]'$




$=\int_{-1}^{\infty}-\frac{1}{3}[-3t]'e^{-3t}dt=-\frac{1}{3}\int_{-1}^{\infty}[-3t]'e^{-3t}dt=\frac{1}{3}\int_{\infty}^{-1}[-3t]'e^{-3t}dt$

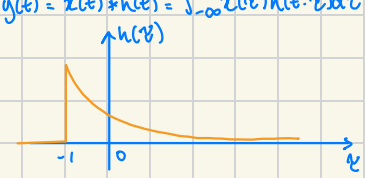
$=\frac{1}{3}\left[e^{-3t}\right]_{\infty}^{-1}=\frac{1}{3}\left[\frac{1}{e^{3(\infty)}}^{0}-e^{-3(-1)}\right]=-\frac{1}{3}e^3<\infty$

$\Rightarrow$ h(t) is stable

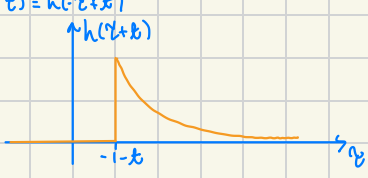
c) $x(t)=u(t)-u(t-5)$

$h(t)=e^{-3t}u(t+1)$

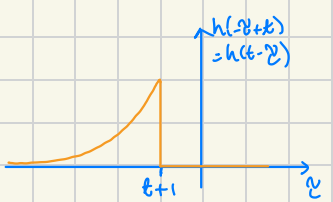
$y(t)=x(t)*h(t)=\int_{-\infty}^{\infty}x(\tau)h(t-\tau)d\tau$, $h(t-\tau)=h(-\tau+t)$

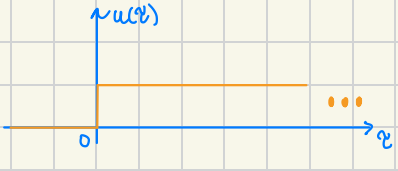


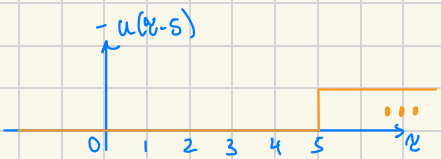

$\Rightarrow$



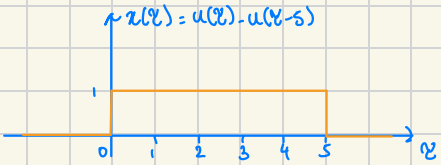

$\Rightarrow$

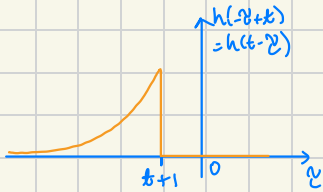


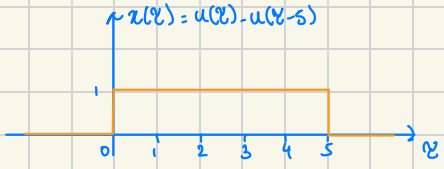



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$h(t) = e^{-3t}u(t+1)$

$h(t-\tau) = e^{-3(t-\tau)}u(t-\tau+1) = e^{-3t}e^{3\tau}u(t-\tau+1)$

for : $t+1 \geq 0 \Rightarrow t \geq -1 \Rightarrow$ no overlap $\Rightarrow$ $\boxed{y(t)=0}$

for : $t+1 \geq 0 \Rightarrow t \geq -1 \Rightarrow y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} \underbrace{(u(\tau)-u(\tau-5))}_{=1} e^{-3t}e^{3\tau} \underbrace{u(t-\tau+1)}_{=1} d\tau$

$t+1 \leq 5$

$t \leq 4$

$= \int_0^{t+1} e^{-3t}e^{3\tau}d\tau = e^{-3t}\int_0^{t+1} e^{3\tau}d\tau$ , $[3\tau]' = 3 \Rightarrow 1 = \frac{1}{3}[3\tau]'$

for : $-1 \leq t \leq 4$

$= e^{-3t}\int_0^{t+1} \frac{1}{3}[3\tau]'e^{3\tau}d\tau = \frac{1}{3}e^{-3t}\int_0^{t+1}[3\tau]'e^{3\tau}d\tau = \frac{1}{3}e^{-3t}\left[e^{3\tau}\right]_0^{t+1}$

$= \frac{1}{3}e^{-3t}\left[e^{3(t+1)} - \underbrace{e^{3(0)}}_{=1}\right] = \frac{1}{3}e^{-3t}\left[e^{3(t+1)} - 1\right]$

$\Rightarrow \boxed{y(t) = \frac{1}{3}e^{-3t}\left[e^{3(t+1)} - 1\right]}$

for : $t+1 > 5 \Rightarrow t > 4 \Rightarrow t > 4$

$y(t) = \int_{-\infty}^{\infty} e^{-3t}e^{3\tau}d\tau = \int_0^5 e^{-3t}e^{3\tau}d\tau = e^{-3t}\int_0^5 e^{3\tau}d\tau = e^{-3t}\int_0^5 \frac{1}{3}[3\tau]'e^{3\tau}d\tau = \frac{1}{3}e^{-3t}\left[e^{3\tau}\right]_0^5 = \frac{1}{3}e^{-3t}\left[e^{3(5)} - e^{3(0)}\right]$

$= \frac{1}{3}e^{-3t}\left[e^{3(5)} - 1\right] = \frac{1}{3}\left(e^{-3t+3(5)} - e^{-3t}\right) = \frac{1}{3}\left(e^{-3(t-5)} - e^{-3t}\right)$

$\Rightarrow \boxed{y(t) = \frac{1}{3}\left(e^{-3(t-5)} - e^{-3t}\right)}$

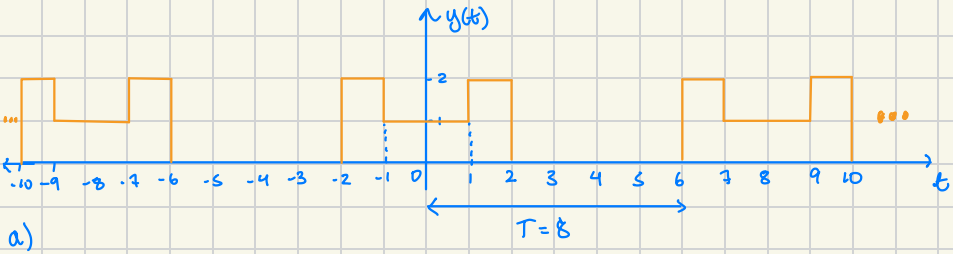
final result :

$$\boxed{y(t) = \begin{cases} 0, & t < -1 \\ \frac{1}{3}e^{-3t}\left[e^{3(t+1)} - 1\right], & -1 \leq t \leq 4 \\ \frac{1}{3}\left(e^{-3(t-5)} - e^{-3t}\right), & t > 4 \end{cases}}$$

Q5:)

a:) Find the Fourier Series coefficients of the periodic signal. You may use the Fourier series properties to do the calculation.

b:) Give the expression of the first-order (that is $a_k e^{jk\omega_0 t} + a_{-k} e^{jk\omega_0 t}$ for $k=1$).




a)

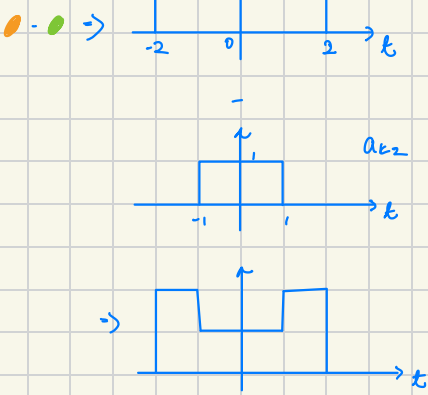
$a_0 = \frac{\text{area}}{T}$ , area $= (1\times2) + (1\times2) + (1\times2) = 2+2+2 = 6$

$a_0 = \frac{6}{8} = \frac{3}{4} \Rightarrow \boxed{a_0 = \frac{3}{4}}$

$a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$ , $k \neq 0$

$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{8} = \frac{\pi}{4} \Rightarrow \boxed{\omega_0 = \frac{\pi}{4}}$




$a_k = a_{k1} - a_{k2}$

$a_{k_1} = \frac{2\sin(k\pi/4(2))}{k\pi} = \frac{2\sin(k\pi/2)}{k\pi}$

$a_{k_2} = \frac{\sin(k\pi/4(1))}{k\pi} = \frac{\sin(k\pi/4)}{k\pi}$

$a_k = \frac{2\sin(k\pi/2)}{k\pi} - \frac{\sin(k\pi/4)}{k\pi}$

$\Rightarrow \boxed{a_k = \frac{2\sin(k\pi/2) - \sin(k\pi/4)}{k\pi}}$

b)

for $k=1 \Rightarrow a_1 = \frac{2\sin(\pi/2) - \sin(\pi/4)}{\pi} = \frac{2 - (1/\sqrt{2})}{\pi} \Rightarrow \boxed{a_1 = \frac{2 - 1/\sqrt{2}}{\pi}}$

($\sin(\pi/2) = 1$, $\sin(\pi/4) = 1/\sqrt{2}$)

$a_k e^{jk\omega_0 t} + a_{-k} e^{jk\omega_0 t}$ , for $k=1$

$a_1 e^{j\omega_0 t} + a_{-1} e^{j\omega_0 t}$ , $a_1 = a_{-1}$ because signal is even and real.

$\Rightarrow \frac{2a_1\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)}{2} = 2a_1 \cos(\omega_0 t)$

$\Rightarrow a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} = \frac{(x)\,2a_1\cos(\omega_0 t)}{(x)} = 2\left(\frac{2-\sqrt{2}}{\pi}\right)\cos\left(\frac{\pi}{4}t\right)$

$\Rightarrow \boxed{a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} = 2\left(\frac{2-\sqrt{2}}{\pi}\right)\cos\left(\frac{\pi}{4}t\right)}$